

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Chemistry
2. Subject Name: Molecular Symmetry and Spectroscopy
L-T-P: 4-0-0 Credits: 4
3. Pre-requisites: NIL
4. Syllabus and reference books:

Syllabus:

Module 1: Introduction to Group Theory: Symmetry and Point groups

- a) Vector Spaces, Matrices and Transformations
- b) Symmetry elements and operations, Multiplication of Symmetry Operators, Groups and Molecular Point Groups, Equivalent, reducible and irreducible representation of Groups, The Great Orthogonality Theorem. Characters: Theorems and Properties, Character table of Point Groups, Direct product, Projection Operators, Symmetry Adapted Linear Combination (SALC), Link between Group theory and Quantum Mechanics.
- c) Chemical Applications: Molecular Vibrations, Normal Coordinates, Selection Rules for IR and Raman Spectroscopy, Characteristic bond frequencies. Group theoretical approach for vanishing integrals and their applications.

Module 2: Introduction to Molecular spectroscopy: absorption, emission and resonance

- a) The basic science of molecular spectroscopy. Wave properties of light, the quantum theory of light. Absorption and emission of light by molecular species. Region of electromagnetic spectrum, emission and absorption spectra, signal to noise ratio and resolving power, width and intensity of spectral transitions/lines. Types of molecular motions and their quantum descriptions
- b) Pure rotational spectra, diatomic rigid rotor molecules, effect of isotope substitution. Rotational spectra of polyatomic molecules.
- c) Vibrational spectroscopy- classical and quantum description of molecular vibrations. Normal modes, character tables. Infrared spectroscopy. Selection rules. The Franck-Condon principle. Analysis of IR spectra. FTIR spectroscopy. The Born-Oppenheimer approximation. Vibration- rotational spectra of diatomic molecules, harmonic oscillator-rigid rotor approximation, anharmonicity effect, Coriolis coupling. Hot bands and overtones Instrumentation used in IR spectroscopy.
- d) Raman spectroscopy: Description of Raman scattering (classical and quantum). Selection rules. Instrumentation used in Raman spectroscopy. Raman spectroscopy at the nanoscale. Introduction to tip-enhanced Raman spectroscopy.
- e) Electronic spectroscopy: Electronic transitions, energy of electronic transitions. Term symbols. UV-VIS spectroscopy. Instrumentation used in UV-VIS spectroscopy. UV-VIS spectroscopy. Instrumentation used in UV-VIS spectroscopy. Emission spectroscopy-fluorescence and phosphorescence. Non-radiative and radiative transitions. Introduction to LASERS and LASER spectroscopy, Technological applications of molecular spectroscopy.
- f) Resonance Spectroscopy:

- i. Classical NMR Spectroscopy: The Vector model, Bloch equations, NMR parameters: pulse width, spectral width, dwell time, number of points, repetition delay, acquisition time, receiver gain, filter width, number of increments, transmitter frequency, transmitter power, decoupling power, Fourier transformation (size and apodization)
- ii. Experimental NMR Spectroscopy: Superconducting magnets, Probes, Shims, Data acquisition, Data processing, Pulse techniques
- iii. NMR Interactions: The NMR chemical shift, Dipolar interaction, Scalar coupling, Relaxation and Dynamic Processes, Introduction to Relaxation: Master equation, Measuring T1 and T2, Relaxation mechanisms, Nuclear Overhauser effect (NOE), Multidimensional NMR Experiments: The pulse sequence, Coherence-transfer, Coherence selection, COSY: Introduction
- iv. Introduction to EPR, and general applications. Inside a typical EPR spectrometer. CIDNP, CIDEP.

Reference Books:

- 1) Chemical Applications of Group Theory, 3rd edition, F. A. Cotton, Wiley, 2008.
- 2) Group Theory and Chemistry, D. M. Bishop, Dover Publications, 1993
- 3) Molecular Symmetry Groups and Chemistry, S. C. Rakshit, Sarat Book House, 2014.
- 4) Molecular Spectroscopy, J. D. Graybeal, McGraw Hill, 1988
- 5) Molecular Spectroscopy, J. L. McHale, CRC Press, 2017.
- 6) Fundamentals of molecular spectroscopy, C. N. Banwell, McGraw-Hill, 1993
- 7) Molecular Spectroscopy by Ira N. Levine, Wiley, 1975

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Vector Spaces, Matrices and Transformations	02
2.	Symmetry elements and operations, Multiplication of Symmetry Operators, Groups and Molecular Point Groups, Equivalent, reducible and irreducible representation of Groups, The Great Orthogonality Theorem	07
3.	Characters: Theorems and Properties, Character table of Point Groups, Direct product, Projection Operators, Symmetry Adapted Linear Combination (SALC)	07
4.	Link between Group theory and Quantum Mechanics	02
5.	Chemical Applications	06
7.	The basic science of molecular spectroscopy	04
8.	Rotational spectroscopy	04
9.	Vibrational spectroscopy	04
10.	Raman spectroscopy	02

11.	Electronic spectroscopy	04
12.	Resonance Spectroscopy	06
Total number of hours		42